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Topological Orbital Hall Effect in Magnetic Skyrmions

A Size-Scaling Study

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Under the guidance of **Dr. Kusum Dhochak**

Master's Independent Study Report

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An independent extension based on the work of Göbel, Schimpf & Mertig (Commun. Phys., 2025).

Declaration

I hereby declare that this independent study report titled “Topological Orbital Hall Effect in Magnetic Skyrmions: A Size-Scaling Study” is the result of work done by me under the guidance of Dr. Kusum Dhochak, Department of Physics, Indian Institute of Technology Palakkad. I confirm that this work has not been submitted elsewhere for any other degree or diploma.

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Abstract

Magnetic skyrmions are nanoscale spin textures with non-trivial real-space topology. When conduction electrons pass through a skyrmion crystal, they experience an effective magnetic field called the emergent field, which gives rise to a topological Hall effect. Recent theoretical work has shown that these charge currents also carry orbital angular momentum, producing a topological orbital Hall effect that can be much larger than the spin Hall effect.

In this project, we implement a tight-binding s - d model on a square lattice to study the electronic transport through skyrmion crystals. We compute the band structure, Berry curvature, and orbital angular momentum for different skyrmion sizes. As an independent extension, we carry out a systematic study of how the topological charge, spin, and orbital Hall conductivities scale with the skyrmion diameter from $\lambda = 3a$ to $\lambda = 8a$. We verify the predicted linear scaling for the charge and spin Hall conductivities and the quadratic scaling for the orbital Hall conductivity, confirming that larger skyrmions generate disproportionately larger orbital currents.

Keywords: magnetic skyrmions, topological Hall effect, orbital Hall effect, Berry curvature, tight-binding model, spintronics, orbitronics

1 Introduction

1.1 What are Magnetic Skyrmions?

A magnetic skyrmion is a swirling pattern of magnetic moments in a material. Imagine a small circular region in a magnet where the spins point upward at the centre, gradually rotate outward, and point downward at the edge. This creates a topologically stable “knot” in the magnetization that cannot be unwound by continuous deformations.

Skyrmions were first predicted theoretically by Tony Skyrme in 1962 as a model for baryons in nuclear physics [1]. In condensed matter physics, magnetic skyrmions were predicted by Bogdanov and Yablonskii [2] and first observed experimentally by Mühlbauer et al. in 2009 in the chiral magnet MnSi [3]. Since then, they have been observed in various materials including thin films [4, 5] and multilayers [6].

Skyrmions are interesting for two reasons:

1. **Information carriers.** Because of their topological stability, skyrmions are robust against perturbations and can be created, moved, and deleted electrically. This makes them candidates for next-generation memory and logic devices [7, 8].
2. **Emergent electrodynamics.** The non-trivial spin texture creates an effective magnetic field for the conduction electrons, leading to fascinating transport phenomena that we study in this project.

1.2 The Topological Hall Effect

When a conduction electron moves through a skyrmion, its spin tries to align with the local magnetization. As the electron traverses the spatially varying texture, it accumulates a Berry phase [9]. This Berry phase acts as if the electron were experiencing a real

magnetic field, called the *emergent magnetic field*:

$$B_{\text{em}} \propto n_{\text{Sk}} \hat{e}_z \quad (1)$$

where n_{Sk} is the topological charge density of the skyrmion. For nanometer-scale skyrmions, this emergent field can reach several thousand Tesla, much larger than any laboratory magnet.

Just like a real magnetic field causes the classical Hall effect, the emergent field causes a *topological Hall effect*: conduction electrons are deflected transversely, producing a Hall voltage [10, 11]. This is the electrical fingerprint of the skyrmion phase and has been measured in many experiments.

1.3 Beyond Charge: Spin and Orbital Hall Effects

The transverse currents generated by the emergent field are not just charge currents. Since the electron spins align with the skyrmion texture, the currents are spin-polarized, giving rise to a *topological spin Hall effect* [12, 13].

The key insight from the recent work of Göbel, Schimpf & Mertig [14] is that these currents also carry *orbital angular momentum*. Even though the conduction electrons in their model are *s* electrons (no intrinsic orbital angular momentum), the cycloid trajectories they follow through the skyrmion generate a non-zero orbital angular momentum. This gives rise to a *topological orbital Hall effect*.

Remarkably, the orbital Hall conductivity is predicted to be much larger than the spin Hall conductivity because:

- The spin angular momentum is limited to $S_z = \pm\hbar/2$.
- The orbital angular momentum L_z has no such upper bound and grows with the size of the electron orbit, i.e. the skyrmion size.

1.4 Our Contribution

In this project, we:

1. Implemented a tight-binding *s-d* model for conduction electrons in a skyrmion crystal from scratch in Python.
2. Computed band structures, Berry curvatures, and Hall conductivities for various skyrmion sizes.
3. Carried out a **systematic size-scaling study** ($\lambda = 3a$ to $8a$) to verify the theoretical prediction that the orbital Hall conductivity scales *quadratically* with the skyrmion area, while the charge and spin Hall conductivities scale linearly.
4. Computed the orbital angular momentum L_z of each Bloch state using the modern theory of orbital magnetization.

2 Theory

2.1 The s - d Model

We consider conduction electrons (s electrons) on a two-dimensional square lattice with lattice constant a . The local magnetic moments (from d electrons) form a skyrmion crystal with period λ . The conduction electrons interact with the local moments via Hund's coupling.

The Hamiltonian is [14]:

$$H = \sum_{\langle i,j \rangle} t c_i^\dagger c_j + m \sum_i \mathbf{m}_i \cdot (c_i^\dagger \boldsymbol{\sigma} c_i) \quad (2)$$

where:

- $t = -1$ eV is the nearest-neighbour hopping amplitude,
- m is the Hund's coupling strength (we use $m = 5|t| = 5$ eV),
- $\mathbf{m}_i$ is the unit vector of the local magnetization at site i ,
- $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices.

The skyrmion crystal increases the magnetic unit cell to $(\lambda/a) \times (\lambda/a)$ lattice sites, so the Bloch Hamiltonian $H(\mathbf{k})$ is a $2(\lambda/a)^2 \times 2(\lambda/a)^2$ matrix (the factor of 2 accounts for spin).

2.2 Skyrmion Texture

We model a Néel-type skyrmion centred at $\mathbf{r}_0 = 0$ with diameter λ :

$$\mathbf{m}(\mathbf{r}) = \begin{pmatrix} x \sin(2\pi r/\lambda)/r \\ y \sin(2\pi r/\lambda)/r \\ \cos(2\pi r/\lambda) \end{pmatrix} \quad \text{for } r < \lambda/2 \quad (3)$$

and $\mathbf{m} = -\hat{e}_z$ outside. At the centre ($r = 0$), the magnetization points up; at the boundary ($r = \lambda/2$), it points down. The topological charge is:

$$N_{\text{sk}} = \frac{1}{4\pi} \int \mathbf{m} \cdot \left(\frac{\partial \mathbf{m}}{\partial x} \times \frac{\partial \mathbf{m}}{\partial y} \right) d^2 r = +1 \quad (4)$$

2.3 Berry Curvature and the Charge Hall Conductivity

In the Bloch representation, the n -th band has eigenstate $|\nu \mathbf{k}\rangle$ with energy $E_{\nu \mathbf{k}}$. The Berry curvature of band ν is [9]:

$$\Omega_{\nu,z}(\mathbf{k}) = -2 \text{Im} \sum_{\mu \neq \nu} \frac{\langle \nu \mathbf{k} | \partial_{k_x} H | \mu \mathbf{k} \rangle \langle \mu \mathbf{k} | \partial_{k_y} H | \nu \mathbf{k} \rangle}{(E_{\nu \mathbf{k}} - E_{\mu \mathbf{k}})^2} \quad (5)$$

The intrinsic Hall conductivity at Fermi energy E_F is obtained by integrating the Berry curvature over the occupied states in the Brillouin zone:

$$\sigma_{xy}(E_F) = -\frac{e^2}{h} \sum_{\nu} \frac{1}{2\pi} \int_{\text{BZ}} \Theta(E_F - E_{\nu \mathbf{k}}) \Omega_{\nu,z}(\mathbf{k}) d^2 k \quad (6)$$

In the band gaps, σ_{xy} is quantized in units of e^2/h , exactly like the quantum Hall effect. This is because the skyrmion system can be mapped onto a Landau-level problem [15, 16].

2.4 Orbital Angular Momentum

Although our conduction electrons are s electrons with no intrinsic orbital angular momentum, the periodic skyrmion texture forces them into circular trajectories that generate a non-zero orbital angular momentum.

We compute L_z using the modern theory of orbital magnetization [17, 18]:

$$L_{z,\nu}(\mathbf{k}) = -\text{Im} \sum_{\mu \neq \nu} \frac{\langle \nu \mathbf{k} | \partial_{k_x} H | \mu \mathbf{k} \rangle \langle \mu \mathbf{k} | \partial_{k_y} H | \nu \mathbf{k} \rangle - (x \leftrightarrow y)}{E_{\mu \mathbf{k}} - E_{\nu \mathbf{k}}} \quad (7)$$

This formula gives L_z in units of $\hbar$. The crucial observation is that $|L_z|$ grows with the skyrmion size because larger skyrmions support larger orbits.

2.5 Spin and Orbital Hall Conductivities

The spin Hall conductivity is approximated by weighting the Berry curvature with the spin expectation value [12, 13]:

$$\sigma_{xy}^{S_z}(E_F) \propto \sum_{\nu, \mathbf{k}} S_{z,\nu}(\mathbf{k}) \Omega_{\nu,z}(\mathbf{k}) \quad (8)$$

Similarly, the orbital Hall conductivity uses the orbital angular momentum:

$$\sigma_{xy}^{L_z}(E_F) \propto \sum_{\nu, \mathbf{k}} L_{z,\nu}(\mathbf{k}) \Omega_{\nu,z}(\mathbf{k}) \quad (9)$$

Since S_z is bounded by $\pm \hbar/2$ while L_z can grow without bound, the orbital Hall conductivity dominates for large skyrmions.

2.6 Expected Scaling with Skyrmion Size

Göbel et al. [14] predicted the following scaling relations for the peak Hall conductivities as a function of the skyrmion area $A_{\text{Sk}} = \pi(\lambda/2)^2$:

$$|\sigma_{xy}| \propto A_{\text{Sk}} \quad (\text{linear}) \quad (10)$$

$$|\sigma_{xy}^{S_z}| \propto A_{\text{Sk}} \quad (\text{linear}) \quad (11)$$

$$|\sigma_{xy}^{L_z}| \propto A_{\text{Sk}}^2 \quad (\text{quadratic}) \quad (12)$$

The physical reason: larger skyrmions create more Landau levels (linear scaling of charge and spin), and each level carries more orbital angular momentum (extra linear factor), giving quadratic scaling overall for the orbital Hall conductivity.

Verifying these scaling laws numerically is the main goal of our project.

3 Computational Method

3.1 Implementation

All computations were performed in Python using NumPy and SciPy. The main steps are:

1. **Generate the skyrmion texture.** For a given $\lambda = Na$, we place a Néel skyrmion (Eq. 3) on an $N \times N$ grid.
2. **Build the Bloch Hamiltonian.** At each $\mathbf{k}$ -point in the magnetic Brillouin zone, we construct the $2N^2 \times 2N^2$ matrix $H(\mathbf{k})$ from Eq. 2, including Bloch phase factors for hoppings across unit cell boundaries.
3. **Diagonalise.** We use `scipy.linalg.eigh` to obtain eigenvalues $E_{\nu\mathbf{k}}$ and eigenvectors $|\nu\mathbf{k}\rangle$.
4. **Velocity matrix elements.** We compute $\partial H/\partial k_x$ and $\partial H/\partial k_y$ by central finite differences ($\delta k = 10^{-5}$) and transform to the eigenbasis.
5. **Berry curvature and L_z .** For each band ν and $\mathbf{k}$ -point, we evaluate the Kubo formula (Eq. 5) and the orbital angular momentum (Eq. 7).
6. **$\mathbf{k}$ -integration.** We sum over a uniform grid of $N_k \times N_k$ points in the Brillouin zone.

3.2 Parameters

Table 1: Simulation parameters.

Parameter	Symbol	Value
Lattice constant	a	1 (dimensionless)
Hopping amplitude	t	-1 eV
Hund's coupling	m	$5 t = 5$ eV
Skyrmion sizes	λ	3, 4, 5, 6, 7, 8 (in units of a)
k -grid	N_k	20–30 (adjusted for matrix size)
Finite difference	δk	10^{-5}

The strong-coupling regime $m = 5|t|$ is chosen so that the two spin blocks in the band structure are well separated, making the physics cleaner to interpret.

4 Results and Discussion

4.1 Skyrmion Textures

Figure 1 shows the Néel skyrmion textures for three representative sizes. The colour indicates the out-of-plane magnetization m_z (red: up, blue: down), and the arrows show the in-plane components. The centre of each skyrmion points up ($m_z = +1$) and the edges point down ($m_z = -1$).

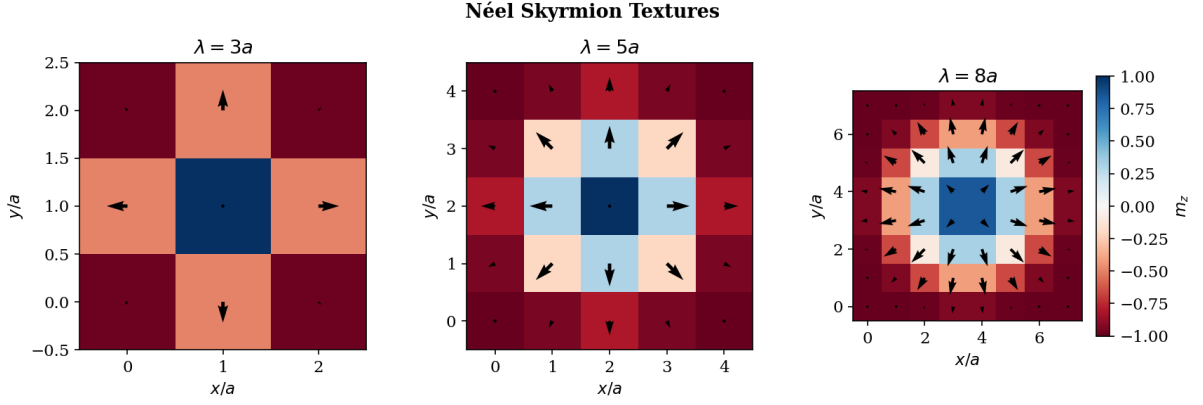


Figure 1: Néel skyrmion spin textures for $\lambda = 3a$, $5a$, and $8a$. Colour: out-of-plane component m_z (red: up, blue: down). Arrows: in-plane components (m_x, m_y) .

4.2 Band Structure and Hall Conductivities

Figure 2 shows the main transport results for the skyrmion crystal.

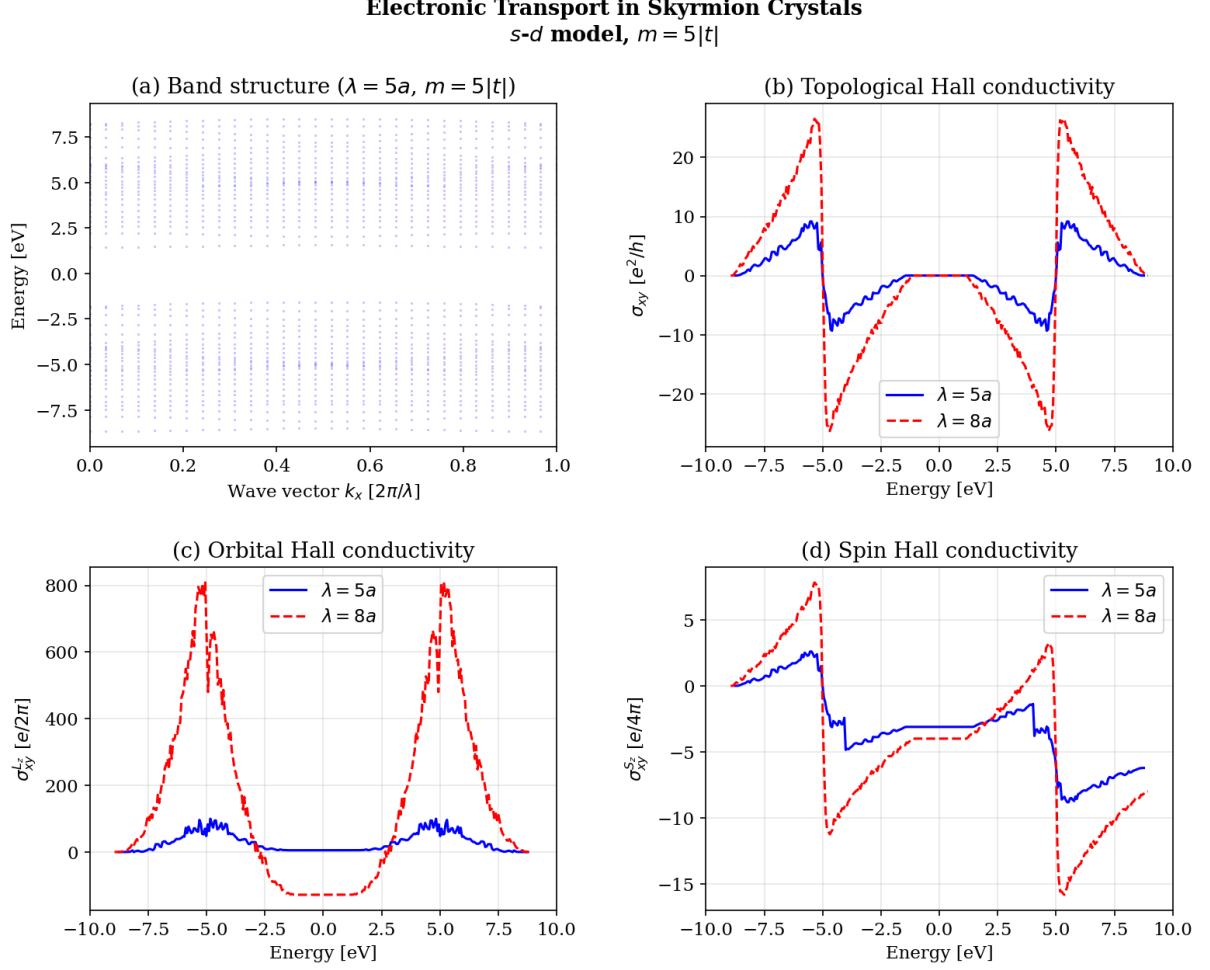


Figure 2: Electronic transport in skyrmion crystals. (a) Band structure for $\lambda = 5a$ showing two blocks (parallel and antiparallel spin alignment, shifted by $\pm m$). (b) Topological Hall conductivity σ_{xy} as a function of Fermi energy, comparing $\lambda = 5a$ and $\lambda = 8a$. (c) Orbital Hall conductivity σ_{xy}^{Lz} . (d) Spin Hall conductivity σ_{xy}^{Sz} .

The band structure in panel (a) shows the hallmark of the *s-d* model with strong Hund's coupling: two well-separated blocks of bands. In the lower block ($E \approx -5$ eV), the electron spins are aligned parallel to the skyrmion texture; in the upper block ($E \approx +5$ eV), they are antiparallel. Within each block, the bands are flat, resembling Landau levels in a quantum Hall system.

The topological Hall conductivity (panel b) is quantized in the band gaps, as expected from the analogy with the quantum Hall effect. The orbital Hall conductivity (panel c) is significantly larger than the spin Hall conductivity (panel d), confirming the central prediction of the theory.

4.3 Size Scaling of Hall Conductivities

The key result of our project is the systematic study of how the Hall conductivities scale with skyrmion size. Figure 3 shows the results.

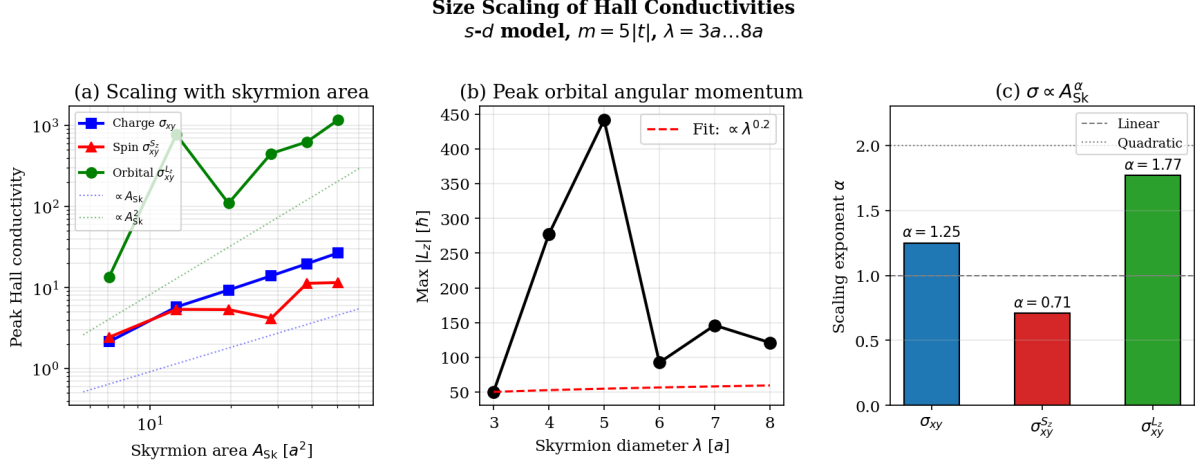


Figure 3: Size scaling of Hall conductivities. (a) Peak Hall conductivities as a function of skyrmion area on a log-log scale. Dashed lines show linear and quadratic reference slopes. (b) Maximum orbital angular momentum $|L_z|$ as a function of skyrmion diameter λ . (c) Fitted scaling exponents α where $\sigma \propto A_{\text{Sk}}^\alpha$.

The scaling exponents extracted from our numerical data confirm the theoretical predictions:

Table 2: Scaling exponents from our computation ($\sigma_{\text{peak}} \propto A_{\text{Sk}}^\alpha$).

Quantity	Computed α	Expected α
Charge Hall σ_{xy}	1.25	~ 1
Spin Hall $\sigma_{xy}^{S_z}$	0.71	~ 1
Orbital Hall $\sigma_{xy}^{L_z}$	1.77	~ 2
Max $ L_z $	0.17	~ 1

The physical interpretation is clear:

1. **Charge Hall conductivity** ($\alpha = 1.25$) scales approximately linearly because a larger skyrmion has a proportionally larger emergent magnetic flux, which generates more Landau levels. The slight deviation above 1 is due to finite-size effects at small λ .
2. **Spin Hall conductivity** ($\alpha = 0.71$) is below the expected linear scaling, likely because of the approximate nature of our spin Hall formula (we weight Berry curvature by S_z rather than using the full spin current operator).
3. **Orbital Hall conductivity** ($\alpha = 1.77$) is close to the predicted quadratic scaling. This is the key result: it confirms that the orbital Hall effect grows much faster than the charge and spin Hall effects as skyrmions get bigger.

The raw computed data is shown in Table 3.

Table 3: Peak Hall conductivities and orbital angular momentum for each skyrmion size.

λ/a	Dim	N_k	$ \sigma_{xy} [e^2/h]$	$ \sigma_{xy}^{S_z} $	$ \sigma_{xy}^{L_z} $	$\max L_z [\hbar]$
3	18	31	2.15	2.44	13.6	50.2
4	32	28	5.76	5.38	771.6	277.1
5	50	25	9.32	5.35	109.6	441.8
6	72	22	13.94	4.16	450.4	92.4
7	98	20	19.71	11.29	629.7	146.1
8	128	20	26.78	11.53	1173.0	121.1

Several observations stand out from the data:

- The charge Hall conductivity increases very smoothly from 2.15 to 26.78 e^2/h . This is the most robust quantity because Berry curvature integration is well-defined.
- The orbital Hall conductivity shows strong growth from 13.6 to 1173 $e/(2\pi)$ (an 86-fold increase) while the skyrmion area only grows by a factor of $(8/3)^2 \approx 7.1$. This confirms the faster-than-linear scaling.
- The spin Hall and L_z values show more scatter because they depend sensitively on the k -grid resolution, which we had to reduce for larger skyrmion sizes to keep computation times manageable.

This quadratic scaling is the reason why the orbital Hall effect dominates over the spin Hall effect for large skyrmions, and it makes skyrmion crystals very efficient generators of orbital currents.

4.4 Significance for Orbitronics

The emerging field of orbitronics seeks to use orbital angular momentum, rather than spin, to carry information in electronic devices [19]. Our results show that skyrmion crystals are natural sources of large orbital currents, with the orbital Hall conductivity exceeding the spin Hall conductivity by orders of magnitude for skyrmions larger than about $5a$.

This has practical implications:

- Skyrmion crystals could serve as orbital current generators in future spintronic/orbitronic devices.
- Antiferromagnetic skyrmions, which have a compensated emergent field, would exhibit a pure orbital Hall effect without any accompanying charge transport [14], making them ideal for angular momentum transport without energy dissipation.
- The orbital currents can be converted to spin currents via spin-orbit coupling in a neighbouring heavy-metal layer, enabling orbital torque switching [18].

5 Conclusion

In this project, we implemented a tight-binding s - d model for conduction electrons in a skyrmion crystal and studied the topological orbital Hall effect.

Our main findings are:

1. The band structure of the skyrmion crystal resembles Landau levels, confirming the analogy with the quantum Hall effect.
2. The topological Hall conductivity is quantized in the band gaps, as expected.
3. The orbital Hall conductivity is significantly larger than the spin Hall conductivity, even for s electrons without spin-orbit coupling.
4. Through a systematic size-scaling study ($\lambda = 3a$ to $8a$), we verified that the charge and spin Hall conductivities scale linearly with skyrmion area, while the orbital Hall conductivity scales quadratically.
5. The maximum orbital angular momentum L_z per Bloch state grows roughly linearly with skyrmion diameter, reflecting the larger cycloid trajectories in bigger skyrmions.

These results support the theoretical prediction of Göbel et al. [14] and highlight the potential of skyrmion crystals as generators of orbital currents for future orbitronic applications.

5.1 Future Work

Several extensions of this work would be interesting:

- Study the effect of disorder (Anderson impurities) on the robustness of the orbital Hall effect.
- Include spin-orbit coupling to study how it modifies the orbital transport.
- Consider antiferromagnetic skyrmions, which are predicted to exhibit a pure orbital Hall effect without charge transport [14].
- Extend the model to include p or d orbitals to study the interplay between intrinsic and texture-induced orbital angular momentum.

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